

 	Republic of the Philippines Department of Transportation
	Document No: <u>DOTr-HRDD-Forms-004</u> Rev. No.: <u>001</u> Effective Date: <u>03 August 2022</u>
	RE-ENTRY ACTION PLAN (REAP) FORM

INSTRUCTIONS: This form is used in preparing a proposed Re-entry Action Plan in applying to a Scholarship Program. Duly accomplished and signed forms will be submitted to the HRD Division along with the other documentary requirements.

Title of Scholarship Program:	JDS 24th Batch Scholarship Program
Name of the Development Partner	Japan International Cooperation Center (JICE)

I. REAP BACKGROUND AND INFORMATION	
a. REAP Title	Building Capacity for Sustainable Rail Transport: Integrating ISO 50001 Energy Management into the Philippine Railways Institute Framework
b. REAP Objective	To build sector-wide capacity for sustainable rail transport, the Philippine Railways Institute (PRI) will develop and integrate ISO 50001-aligned training modules that equip railway personnel across the sector with competencies to achieve measurable improvements in energy performance, targeting a minimum 5% reduction in energy intensity and 7% reduction in energy-related operational expenditures within 36 months of program rollout.
c. Problem/Opportunity	The Philippine Railways Institute (PRI), under the Department of Transportation (DOTr), plays a vital role in developing the technical and managerial capabilities of railway professionals in the country. As the Philippines moves toward modern and electrified railway systems—led by major projects such as the North-South Commuter Railway (NSCR), Metro Manila Subway (MMSP), and MRT-7—PRI now faces the crucial task of modernizing its training programs to meet the demands of advanced technology and sustainability. One major challenge is the limited integration of Energy Management System (EMS) concepts in existing training and operations. Although the new electrified lines feature energy-efficient technologies like regenerative braking and onboard energy storage, most of PRI's current competency-based programs still focus on traditional areas such as operations, maintenance, and safety. This creates a gap in skills related to energy flow analysis, traction energy optimization, and the practical use of EMS tools. Without this knowledge, railway engineers and operators have fewer opportunities to apply data-driven approaches that could improve energy efficiency and reduce costs. At the same time, this challenge offers a valuable opportunity for PRI to strengthen its role as a center of excellence in sustainable railway energy management. By integrating EMS-related modules into existing courses—such as those for train driving, power supply and distribution,

	and rolling stock maintenance – PRI can help railway personnel gain the expertise needed to manage and optimize energy use in modern electric train systems. The importance of this initiative is reinforced by current data. The Philippine Statistics Authority (2023) reports that transportation accounts for nearly 35% of total energy consumption, highlighting the sector’s critical role in national energy demand. Furthermore, studies from the Japan International Cooperation Agency (JICA) and the Asian Development Bank (ADB) show that improving energy efficiency in public transport – especially railways – can greatly reduce both energy use and greenhouse gas emissions. Japan’s experience, where railway companies such as JR East and Tokyo Metro achieved energy savings of 15–30% through EMS implementation and regenerative braking optimization, demonstrates this potential. Developing and institutionalizing EMS-focused training within PRI supports national goals under AmBisyon Natin 2040 and the Philippine Development Plan, both of which call for modern, efficient, and sustainable transport systems. It also contributes to global commitments such as SDG 7 (Affordable and Clean Energy), SDG 9 (Industry, Innovation, and Infrastructure), and SDG 11 (Sustainable Cities and Communities). Through this initiative, PRI can address existing skill gaps while leading the country’s transition toward a cleaner, smarter, and more energy-efficient railway sector – supporting both national development and environmental sustainability.
d. Description in relation to the Department i. Organizational Outcome	The proposed REAP, titled “Building Capacity for Sustainable Rail Transport: Integrating ISO 50001 Energy Management into the Philippine Railways Institute Framework,” seeks to enhance the Philippine Railways Institute’s (PRI) institutional and technical capacity under the Department of Transportation (DOTr). The initiative focuses on embedding ISO 50001-based Energy Management System (EMS) principles into PRI’s training and organizational framework to promote sustainability, efficiency, and continuous improvement in railway operations. By introducing these principles, the project aims to improve service delivery across major rail networks such as the NSCR, PNR, LRTA, and MRT-3, enabling more energy-efficient, reliable, and cost-effective train operations that reflect global best practices. To ensure long-term implementation, the project will establish an Energy Management and Sustainability Unit (EMSU) within PRI to oversee EMS-related activities, facilitate collaboration with partner agencies, and ensure the consistency of energy management initiatives. Alongside this, a Training Management and Monitoring System (TMMS) will be developed to integrate EMS performance indicators into training evaluation and monitoring. This system will help measure learning outcomes, track progress, and guide continuous improvement in line with ISO 50001 standards and emerging railway technologies. The REAP also supports policy development by producing technical recommendations and policy inputs for DOTr based on the pilot implementation of the EMS-focused training. These outputs will help standardize EMS practices and energy performance monitoring across future railway projects. At the core of the initiative is capacity building – the creation of an EMS-centered training module covering topics such as regenerative braking, traction energy optimization, and energy storage

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	integration. These lessons will be incorporated into existing PRI courses on train driving, power supply and distribution, and rolling stock maintenance. Through this approach, PRI will equip railway professionals with the skills to apply energy management practices in real operations while fostering a culture of sustainability and innovation. Overall, this REAP strengthens PRI's role as a center of excellence in sustainable railway development and supports the Philippines' broader goal of achieving a modern, low-carbon, and energy-efficient transport system.
ii. Target Competency Improvement	Through this study opportunity, I aim to enhance my competencies in energy systems management, transport sustainability, and data-driven infrastructure optimization. These skills are essential for developing and institutionalizing an Energy Management System (EMS) framework suited to the operational needs of the Philippine railway sector. I expect to gain technical proficiency in energy modeling, system simulation, and regenerative braking optimization, as well as deeper knowledge of energy storage integration and performance evaluation in railway systems. Strengthening these competencies will enable me to analyze traction power flow, identify efficiency opportunities, and design strategies that reduce operational energy costs and emissions. In addition, I seek to refine my capabilities in policy analysis, project evaluation, and institutional capacity development, which are vital for embedding EMS knowledge within the Philippine Railways Institute's (PRI) competency-based training framework. These combined skills will allow me to translate technical learning into actionable training materials and policy recommendations, supporting the Department of Transportation's (DOTr) broader goals for sustainable and energy-efficient railway operations.
e. Relation to the Government's Development Plan	The proposed REAP strongly supports the objectives of the Philippine Development Plan (PDP) 2023–2028, particularly in advancing human capital development, improving infrastructure, and promoting climate resilience. Under Chapter 2 – Promote Human and Social Development, specifically Subchapter 2.2 – Improve Education and Lifelong Learning, the project enhances the country's technical-vocational education system by introducing an Energy Management System (EMS) training module tailored for railway personnel. This initiative contributes to the PDP's vision of nurturing a future-ready workforce by promoting lifelong learning, strengthening technical competencies, and encouraging continuous knowledge exchange across the transport sector.

	Aligned with Chapter 12 – Expand and Upgrade Infrastructure, the REAP also supports the development of skilled professionals essential for the ongoing modernization of the Philippine railway network, including major operators such as LRTA, LRM, MRT-7, and PNR. By integrating EMS principles into the Philippine Railways Institute’s (PRI) training framework, the project helps improve energy efficiency and operational performance while contributing to the PDP’s broader goal of providing reliable, safe, and sustainable transport services nationwide. In addition, the initiative reinforces Chapter 15 – Accelerate Climate Action and Strengthen Disaster Resilience by promoting energy-saving technologies such as regenerative braking and power optimization in electric railway systems. These actions support the country’s transition toward a low-carbon transport sector and align with the PDP’s commitment to building an inclusive, innovation-driven, and climate-resilient economy. The REAP also complements the Department of Transportation’s (DOTr) strategic targets, particularly its mission to develop a safe, reliable, and environmentally sustainable transport system supported by a highly competent workforce. By strengthening the institutional capacity of PRI, the project advances the DOTr’s Railway Sector Modernization and Capacity-Building Program, ensuring that Filipino railway professionals are equipped with the technical expertise to manage energy-efficient and sustainable railway operations. Altogether, this initiative contributes meaningfully to national and sectoral goals, linking human resource development, modern infrastructure, and environmental stewardship in support of the country’s sustainable transport agenda.
f. Direct Beneficiary	The immediate beneficiaries of the REAP are the railway professionals and trainees of the Philippine Railways Institute (PRI) under the Department of Transportation (DOTr). This includes engineers, technicians, train drivers, and operations personnel from major railway operators such as the Light Rail Transit Authority (LRTA), Light Rail Manila Corporation (LRMC), Metro Rail Transit Line 7 (MRT-7), and the Philippine National Railways (PNR). Approximately 80–100 personnel are expected to directly benefit during the pilot training phase, with hundreds more gaining from future rollouts as the EMS module becomes institutionalized within the PRI competency framework. The REAP will equip these professionals with specialized knowledge and technical skills in energy management, regenerative braking, and energy storage integration—enhancing their ability to operate and maintain more energy-efficient railway systems. This capacity-building effort will also strengthen PRI’s institutional framework and promote sustainable operations across the national railway network. In the long term, improved energy efficiency, operational reliability, and reduced carbon emissions will not only benefit railway operators but also enhance commuter experience and contribute to the Philippine nation’s pursuit of a sustainable, low-carbon transport system, aligning with national development and climate resilience goals.
g. Gender Equality, Disability, and Social Inclusion (GEDSI)	The REAP promotes inclusive participation in the railway sector by ensuring that women and other marginalized groups such as persons with disabilities (PWDs), indigenous peoples (IPs), and members of the

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	LGBTQ+ community benefit from knowledge-sharing and technical capacity-building activities. Through the Energy Management System (EMS) training and workshops, the project encourages gender-balanced representation among participants, particularly in the technical and operational divisions of agencies like LRTA, LRMC, MRT-7, and PNR. To address gender and social inclusion gaps, the REAP will adopt equitable access policies, ensuring that training schedules, materials, and venues are accessible to all participants regardless of physical ability or social background. Moreover, the project will integrate gender-sensitive perspectives in energy management planning, emphasizing workplace diversity and inclusive innovation. By empowering a diverse group of professionals, the REAP strengthens institutional capacity and promotes fair participation in sustainable railway operations. Ultimately, this contributes to the broader goals of the Philippine Development Plan 2023–2028 and the UN Sustainable Development Goals—particularly SDG 5 (Gender Equality), SDG 8 (Decent Work), and SDG 10 (Reduced Inequalities).
h. Output	• Develop a comprehensive ISO 50001-based EMS Training Module tailored to the Philippine railway sector. • Produce instructional materials, case studies from Japanese best practices, and simulation exercises covering regenerative braking, traction energy optimization, and energy storage integration. • Create an EMS Implementation Framework to guide standard adoption, monitoring, and continuous improvement across railway operators. • Integrate EMS principles into existing PRI courses, including the Fundamental Training Course, Rolling Stock Maintenance, and Capacity Development Training Course on Energy Management of Railway Systems. • Establish a continuous capacity-building program for engineers and technical personnel from LRTA, LRMC, MRT-7, and PNR, promoting practical application of energy management practices. • Strengthen PRI's role as the national center for railway competency development, enhancing technical expertise and fostering a culture of sustainability and innovation.

	- Promote operational efficiency, energy savings, and environmental sustainability aligned with ISO 50001 standards and the Philippine Development Plan (PDP 2023–2028) goals. - Support DOTr’s strategic objectives for a modern, reliable, and energy-efficient railway system while contributing to national human capital development and climate resilience initiatives
II. REAP TIMELINE AND IMPLEMENTATION	
a. Measurability	The success of the REAP will be monitored through a structured evaluation framework based on measurable indicators of training quality, knowledge transfer, and operational impact. Progress will be tracked using both quantitative metrics (e.g., number of participants trained, percentage of modules completed, and post-training assessment scores) and qualitative indicators (e.g., feedback from trainees, trainers, and partner institutions). The first milestone will be the development and validation of the EMS training module within the first six months after returning from Japan. This includes content preparation, peer review, and approval by the Philippine Railways Institute (PRI) Curriculum Committee. The second milestone, scheduled within the next six months, is the pilot implementation of the training with at least two railway operators such as LRTA, LRMC, MRT-7, or PNR. Feedback and performance evaluation will be conducted to assess effectiveness and identify areas for improvement. The final milestone will be the integration of the EMS training into PRI’s official competency framework, ensuring its institutional adoption. Continuous monitoring through follow-up surveys and periodic module updates will sustain relevance and alignment with PRI’s vision of fostering competent, globally competitive, and safety-driven railway professionals.
b. Sustainability	To sustain the gains of the REAP, the Energy Management System (EMS) training topic will be formally integrated into the Philippine Railways Institute’s (PRI) existing training programs such as the Fundamental Training Course and the Capacity Development Training Course. This ensures that EMS concepts—like regenerative braking, traction energy optimization, and energy storage integration—become a regular part of PRI’s learning modules for railway engineers and operators. After the pilot sessions, the training materials will be reviewed and validated through PRI’s internal Quality Management System to make them part of the official competency framework. To maintain long-term impact, a Training-of-Trainers (ToT) approach will be adopted so that PRI instructors can continue delivering and improving the EMS module even after the REAP ends. Periodic reviews with railway operators and energy specialists will help update the content with the latest technologies. Through this process, the REAP’s benefits—energy-efficient practices, technical competence, and sustainable railway operations—will be institutionalized within PRI’s vision of building a globally competitive and dynamic railway workforce.

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c. Predictive Action Steps with Identified Success Indicators		
i. Phase/Stage	ii. Expected Output	iii. Timeline
Phase 1: Project Preparation and Coordination	Approval of REAP plan by PRI management and coordination with DOTr, and partner railway operators (LRTA, LRMC, MRT-7, PNR).	Month 1-2
Phase 2: EMS Training Module Development	Draft EMS training manual and presentation materials covering regenerative braking, traction energy optimization, and energy storage integration	Month 3-6
Phase 3: Validation and Pilot Implementation	Conduct of two pilot training sessions (Fundamental or Capacity Development Courses) and collection of participant feedback.	Month 7-13
Phase 4: Revision and Institutionalization	Finalized EMS module integrated into PRI's competency framework and Quality Management System (QMS).	Month 14-20
Phase 5: Evaluation and Knowledge Dissemination	Post-training evaluation report and presentation of outcomes to DOTr and PRI for policy input.	Month 20-24
d. Target Date to Start	Immediately after completion of the JDS program	
e. Target Date to End	24 months after project commencement	

III. RESOURCES

Budget Item	Description	Estimated Cost
1. Training Module Development	Research, drafting, layout, and printing of EMS training manuals and presentation materials	P150,000
2. Training Implementation (Food)	Two pilot batches × 20 participants × P600 per participant = P24,000	P24,000
3. Monitoring and Evaluation	Post-training assessment, participant surveys, and reporting documentation	P50,000
4. Documentation and Publication	Printing of final training module, dissemination of reports, and promotional materials	P30,000

5. Contingency Fund (10%)	Miscellaneous and unforeseen expenses	₱25,000
Total Estimated Budget: ₱279,000		

IV. COMMENT(S) AND RECOMMENDATION(S)

a. Immediate Superior (Mentor)	
b. Head of the Office/Division/Service	
c. Human Resource Development Division	

Prepared by:



Digitally signed
by Alvin P. Jamora

ALVIN P. JAMORA

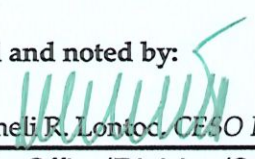
Name of Applicant with signature

Recommended by:

DR. ISRAEL A. RADIAGANDING 

Immediate Supervisor with signature (Mentor)

Approved and noted by:


Usec. Anneli R. Lontoc, CESO I

Head of the Office/Division/Service with signature